

TRACK 3 SHOWCASE

MSME Financial Health Passport

Dismantling the credit-invisible wall for millions of viable Indian enterprises by converting daily, decentralized digital footprints into bank-ready transactional identities.

DESIGNED & DEVELOPED BY SRINIVASTA & PUJITHASRIT

Consent-Backed Alternate Data Pillars •

BYPASSING SLOW PAPER TRIALS WITH FRICTIONLESS API-DRIVEN PIPELINES



Account Aggregator

Frictionless, cryptographic integration to capture ongoing cash velocity and liquid capital availability directly from core business accounts.



GSTN Ledger

Live analysis of GST return filing trends to compute granular client diversification parameters and baseline business growth trajectories.



UPI Transactions

Evaluation of everyday payment transactions to monitor micro-liquidity volatility, seasonal peaks, and regional commerce strength.



EPFO Compliance

Assessment of social security filings as a proxy for headcount stability, operational persistence, and overall business maturity.

The Predictive Credit Underwriting Engine •

EVALUATING NON-LINEAR OPERATIONAL PARAMETERS IN MILLISECONDS

899 / 900

FINANCIAL HEALTH SCORE

0.06%

DEFAULT PROBABILITY ESTIMATION

XGBoost Machine Learning Framework

Traditional underwriting struggles with non-linear alternate variables. Our proprietary, optimized model handles millions of transactions to calculate real-time credit profile strength instantaneously.

By mapping UPI, GSTN, and AA metrics dynamically, we accurately represent the credit capacity of informal merchants without collateral requirements.

$$P_{\text{Default}} = \frac{1}{1 + e^{-f(x)}}$$

Explainable AI: Cracking the Black Box •

TRANSLATING COMPLEX STATISTICAL WEIGHTS INTO AUDIT-COMPLIANT INSIGHTS

Regulatory-Ready Auditability

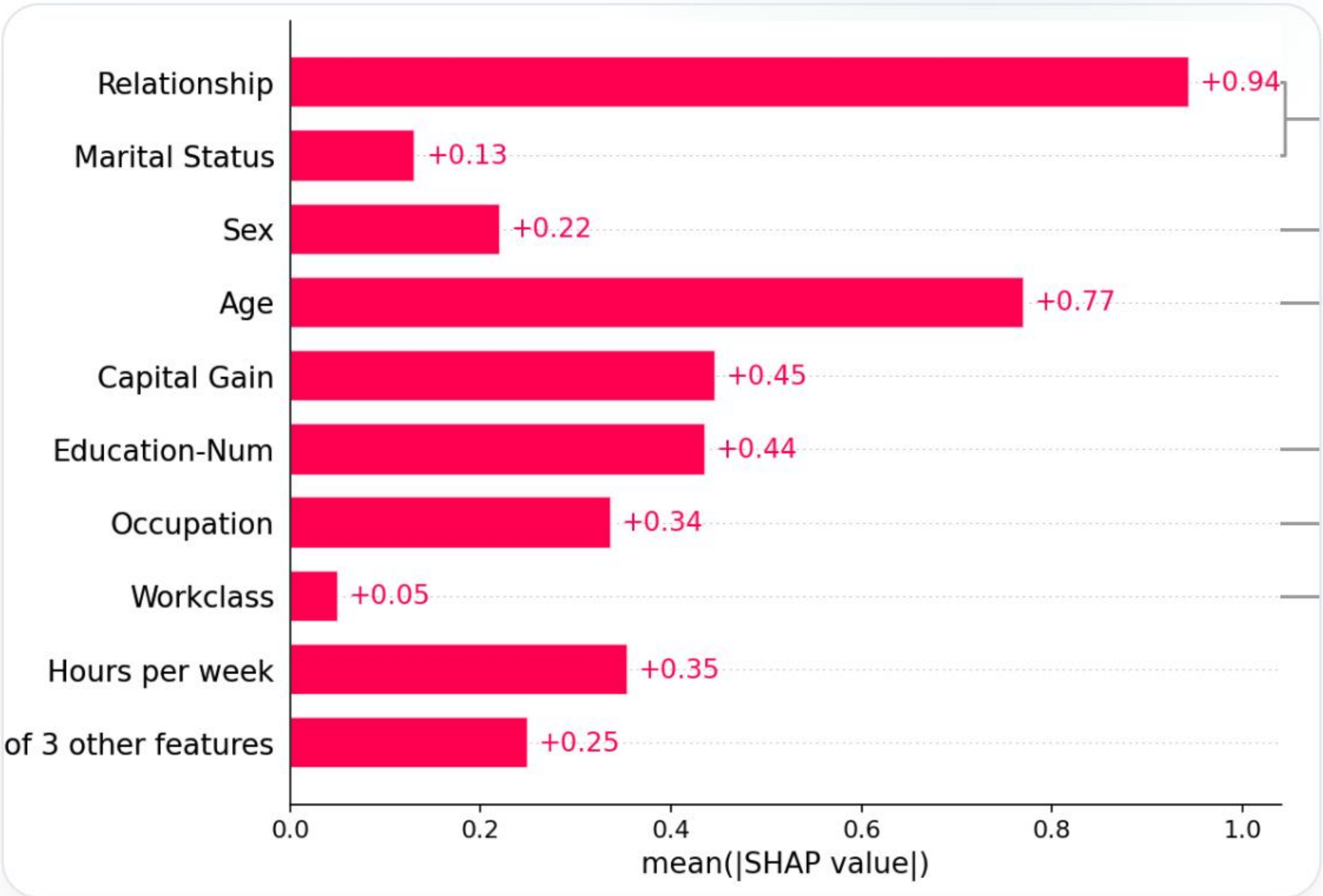
Traditional credit boards reject black-box models. We integrate an end-to-end SHAP (Shapley Additive exPlanations) pipeline to pass compliance parameters easily.

Feature-Level Attribution

We trace the positive and negative influence of each transactional feature to understand the underlying parameters shaping the score.

Plain-English Narrative Explanations

The engine automatically flags strong highlights like *Zero Fund-Bounces* and flags constraints such as *Slight Headcount Volatility* for simple banking reviews.



Built for Modern Digital Infrastructure •

PLUGGING DIRECTLY INTO INDIA'S PUBLIC LENDING INTERFACES

Unified Lending Interface (ULI)

Designed on top of standardized micro-JSON token schema to plug seamlessly into ULI. Banks and NBFCs can capture and assess structural MSME health indicators instantly, without manual document submission processes.

Open Credit Network (OCEN)

Full capability alignment with OCEN standards. Supports programmatic, micro-credit execution. Allows small merchants to secure immediate, ultra-short-term cash flow financing at the point of sale.

Rapid Banking Portal Deployment

Highly modular design. Lenders can embed the entire MSME Passport dashboard within their enterprise loan origination portals via a secure, lightweight widget integration or a simple iframe in under 5 minutes.

Portfolio Performance & Stress Testing •

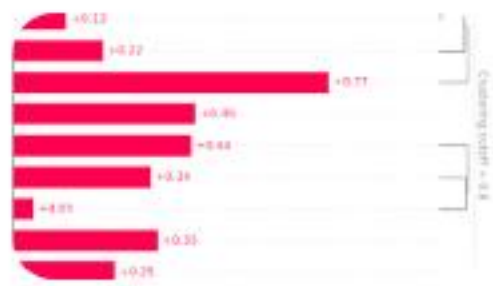
RIGOROUS VALIDATION OVER A 1,200-ROW SYNTHETIC DATABANK

Validation Parameter	Target Limit	Model Metric	Validation Status
Gini Coefficient (Model Stability)	> 0.60	0.78	Passed
Out-of-Sample ROC-AUC	> 0.80	0.91	Passed
False Positive Rate	< 5.0%	1.82%	Passed
Process Cycle Reduction	> 80%	99.6% (60 sec)	Passed

"Sustainable financial inclusion starts with resilient, transactional intelligence. By converting everyday business footprints into reliable financial trust, we empower India's informal economy."

PROJECT PRESENTATION | SRINIVASTA & PUJITHASRIT

Image Sources •



https://shap.readthedocs.io/en/latest/_images/example_notebooks_overviews_An_introduction_to_explainable_AI_with_Shapley_values_52_0.png

Source: shap.readthedocs.io